

Day 16 : Firebase Authentication

1. Introduction to Authentication in firebase

Firebase Authentication is a backend service provided by Google through Firebase that allows users to securely sign in to your Flutter app using email/password, Google, phone number, and other providers without building your own authentication server.

2. why we use Firebase Authentication?

- Secure and easy to implement.
- No need to create your own backend.
- Supports multiple sign-in methods.
- Automatically manages user sessions.
- Works on Android, iOS, and Web.

3. Setup Firebase in Flutter

Step 1: Create a Firebase Project

1. Go to: [Firebase Console](#)
2. Click **Create Project**.
3. Enter project name.
4. Disable/enable Analytics as required.
5. Create the project.

Step 2 : Add Android App

1. Click Add App → Android.
2. Enter:

Android package name

(com.example.myapp)
3. Register app.
4. Download google-services.json.
5. Place it inside:

android/app/google-services.json

Step 3 : Add iOS App (Optional)

1. Click Add App → iOS.
2. Enter Bundle ID.
3. Download: GoogleService-Info.plist

Step 4 : FlutterFire CLI Setup

Install CLI: `dart pub global activate flutterfire_cli`

Login: `firebase login`

Configure Firebase: `flutterfire configure`

Step 5 : Add Dependencies

```
dependencies:  
  firebase_core: ^latest  
  firebase_auth: ^latest  
  google_sign_in: ^latest
```

Install Packages : `flutter pub get`

Step 6 : Initialize Firebase

```
import 'package:firebase_core/firebase_core.dart';  
import 'firebase_options.dart';  
  
void main() async {  
  WidgetsFlutterBinding.ensureInitialized();  
  
  await Firebase.initializeApp(  
    options: DefaultFirebaseOptions.currentPlatform,  
  );  
  
  runApp(MyApp());  
}
```

Step 7. Email/Password Authentication

Enable it in:

Firebase Console → Authentication → Sign-in Method → Email/Password → Enable

Woohoo!

Firebase CLI Login Successful

You are logged in to the Firebase Command-Line interface. You can immediately close this window and continue using the CLI.

i To change your preferences at any time, run 'firebase logout' and 'firebase login' again.

Visit this URL on this device to log in:

https://accounts.google.com/o/oauth2/auth?client_id=563584335869-fqrhgmd47bqnekij5i8b5pr03ho849e6.apps.googleusercontent.com&scope=email%20openid%20https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcloudplatformprojects.readonly%20https%3A%2F%2Fwww.googleapis.com%2Fauth%2Ffirebase%20https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcloud-platform&response_type=code&state=657562025&redirect_uri=http%3A%2F%2Flocalhost%3A9005

Waiting for authentication...

+ Success! Logged in as snehatumaskar@gmail.com

+ You can now use the 'firebase' or 'npm' commands!

~ For more help see <https://firebase.google.com/docs/cli/>

>
> |

PROBLEMS 129 OUTPUT DEBUG CONSOLE TERMINAL PORTS

i Registered a new Firebase web app on Firebase project rise-futureme.

i Firebase windows app rise (windows) is not registered on Firebase project rise-futureme.

i Registered a new Firebase windows app on Firebase project rise-futureme.

Firebase configuration file lib\firebase_options.dart generated successfully with the following Firebase apps:

Platform Firebase App Id

web	1:1041347841229:web:d9c15bea0b103f09336d84
android	1:1041347841229:android:fd43d53fd903df68336d84
ios	1:1041347841229:ios:9db7ae31662a76da336d84
windows	1:1041347841229:web:703af45f6811aff2336d84

Learn more about using this file and next steps from the documentation:

> <https://firebase.google.com/docs/flutter/setup>

PS C:\Users\Sneha\OneDrive\Desktop\LevelUp> |